

05115130 SCMO (June 2026 Class)

Name (in English): _____

Assignment (20%)

Due: Sat 13 June (5:10 PM)

Student ID: _____

Question 1 (20 marks)

a) Consider the optimisation problem

$$\begin{aligned} \text{Minimise} \quad & f(x_1, x_2) = x_1^2 + x_2^2 + x_1x_2 - 5x_2 \\ \text{subject to} \quad & x_1 + 2x_2 = 5, \\ & 2x_1 - x_2 = 10, \\ & x_1 + x_2 \leq 4. \end{aligned}$$

- i) Write down the Lagrangian formula $L(x_1, x_2, \lambda_1, \lambda_2, \mu)$. (1 mark)
- ii) Determine all constraints of $L(x_1, x_2, \lambda_1, \lambda_2, \mu)$. (5 marks)
- iii) Find the values of x_1 and x_2 . (4 marks)

- b) Use the KKT conditions to find and classify the extrema for the following problem.
For inequalities, rewrite them in the form $g(x) \leq 0$ if necessary. (10 marks)

$$\begin{array}{ll}\text{Maximise} & f(x, y) = x^2y \\ \text{subject to} & x + y - 1 = 0, \\ & -x \leq 0.\end{array}$$

Question 2 (20 marks)

An emergency response agency must strategically choose optimal distribution center locations from five candidate facilities (F_1 , F_2 , F_3 , F_4 , and F_5) to distribute a single commodity (vaccines) to seven regional clinics (C_1 through C_7). Given the following information, formulate the optimisation model of the problem.

Table 1: Facility Transportation Costs, Capacities, and Fixed Costs

Facility	Transportation cost (\$)							Capacity (units)	Fixed cost (\$)
	C1	C2	C3	C4	C5	C6	C7		
F1	3	2	4	5	6	4	3	500	1200
F2	4	3	4	2	5	3	4	600	1400
F3	3	4	3	5	4	2	3	550	1300
F4	2	3	5	4	3	5	4	450	1000
F5	5	3	4	3	2	3	5	600	1500
Demand	200	150	180	170	130	140	160		

You are provided with the following notation for use in formulating the model:

- K_j : capacity of facility $j \in V_1 = \{F_1, F_2, \dots, F_5\}$
- F_j : fixed cost to open facility j
- $z_j \in \{0, 1\}$: binary variable, which is 1 if facility j is opened, 0 otherwise
- D_k : demand at clinic $k \in V_2 = \{C_1, C_2, \dots, C_7\}$
- $y_{jk} \geq 0$: quantity shipped from facility j to clinic k
- c_{jk} : transportation cost per unit from facility j to clinic k

a) Define the decision variables required for this optimisation problem. (5 marks)

- b) Formulate the objective function and briefly explain the objective and all the terms
(9 marks)

- c) Write down all the constraints of the model, categorised as follows: (6 marks)

- i) Demand Satisfaction: Ensure that each clinic receives its required quantity.
- ii) Facility Capacity: Ensure that no facility exceeds its shipping capacity, and shipments only occur from open facilities.
- iii) Binary and Non-Negativity: Enforce binary status for facility openings and non-negativity for shipment quantities.

Question 3 (20 points)

A small logistics company is planning how to meet delivery demand for the next **2 months**. They are evaluating two options to cover vehicle capacity:

- **Option 1 – Rent on demand:** Pay for vans as needed from a daily rental company.
- **Option 2 – Monthly lease:** Lease 8 vans for 2 months, and rent extras when needed.

Information:

- Each van can handle **1,000 packages per month**.
 - Expected base demand: **8,000 packages/month**.
 - Demand may rise 20% or fall 20% with the probabilities 0.2 and 0.3, respectively, or remain the same with a probability of 0.5.
 - Rental cost: \$700 per van per month.
 - Lease cost: \$600 per van per month (2-month contract).
 - Revenue: \$1 per package.
 - Monthly discount rate: $k = 10\%$
 - Assume demand is **independent across months**.
- (a) Draw a decision tree diagram to represent the uncertainty of the demand. Include all relevant information such as the demand for each node and probabilities. (5 marks)

- (b) Use the following tables to calculate revenue, costs, and profits for option 1 under different demand scenarios. (10 marks)

Period 2 Demand	Revenue	Cost	Profit
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Period 1 Demand	EMV	PV	Revenue	Cost	Profit
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Period 0 Demand	EMV	PV	Revenue	Cost	Profit
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- (c) If the profit of option 2 is \$6430, which option should the company choose? (5 marks)

Question 4 (20 points)

You are the inventory analyst for a distribution centre that tracks weekly product demand. You are tasked with using two different forecasting methods to predict demand for the upcoming week and evaluate their performance.

Week	1	2	3	4	5	6	7
Demand (Units)	120	130	125	135	140	150	145

- (a) Fill in the following table and determine the regression equation using the least squares method and use the equation to forecast demand for Week 8. (8 marks)

Week X	Demand Y	X^2	XY	$\hat{Y}$	$ Y - \hat{Y} $
1					
2					
3					
4					
5					
6					
7					
Σ					

- (b) Using **Exponential Smoothing (SES)** with smoothing constant $\alpha = 0.4$ and initial forecast of 120, fill in the following table and compute the forecast for Week 8. (8 marks)

Week X	Demand Y	$\hat{Y}$	$ Y - \hat{Y} $
1			
2			
3			
4			
5			
6			
7			
Σ			

- (c) Compute the **mean absolute error (MAE)** for Weeks 1–7 using linear regression and SES methods. Use the following formula for absolute error: (2 marks)

$$MAE = \frac{1}{n} \sum_{i=1}^n |\text{Actual Demand} - \text{Predicted Demand}|$$

- (d) Which method would you recommend for forecasting in this case, and why? (2 marks)

Question 5 (20 marks)

BookNest, a bookstore chain, manages 30,000 books per year for resale. The bookstore pays a fixed delivery fee of **\$500 per order**, regardless of order size. The book supplier offers an **all-unit quantity discount** as follows:

- \$10.00 per book for orders less than 15,000 units
- \$9.70 per book for orders between 15,000 and 30,000 units
- \$9.40 per book for orders greater than 30,000 units

BookNest incurs a **holding cost of 25% per unit price annually**.

- (a) Using the EOQ model with discount, fill in the following table and answer the question in part (b)-(d). (10 marks)

Tier	Order Range	Price (C_i)	Holding Cost (h_i)	Break point (q_i)	EOQ Q	Adjusted Q	Total Cost (TC)
1							
2							
3							

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- (b) What is the optimal order quantity for BookNest? (2 marks)
- (c) What is the annual total cost under this policy? (2 marks)
- (d) What is the cycle inventory maintained by BookNest under the optimal policy? (2 marks)
- (e) How does the cycle inventory compare if the supplier instead charges a flat price of \$9.50 per book with no quantity discounts? (4 marks)